College of Science and Technology

Royal University of Bhutan

Computing Technologies Department

Bachelor of Engineering in Information Technology

02230148

Integrated System

CTE412

LAB REPORT

Autumn Semester 2026

Task 1

1. Create a kind cluster named cte412-<sid> with one control-plane node and two worker nodes. Host port 30090, bound to 127.0.0.1 only, must reach node port 30090 on the cluster.

Kind-cluster.yaml file

```yaml
kind: Cluster
apiVersion: kind.x-k8s.io/v1alpha4
name: cte412-02230148
nodes:
  - role: control-plane
    extraPortMappings:
      - containerPort: 30090 # NodePort inside the kind node
        hostPort: 30090 # published on the Windows host
        listenAddress: "127.0.0.1" # Loopback only, per brief
        protocol: TCP
  - role: worker # notice-board pods
  - role: worker # pod spread + multi-pod /version responses
```

Commands:

- kind create cluster: Tells the KinD CLI to spin up a new local Kubernetes cluster
- --name cte412-02230148: sets the name of the cluster to cte412-02230148.
- --config cluster/kind-cluster.yaml: Reads topology and settings from kind-cluster.yaml

```
PS D:\022030148(Sonam))\SEM VII\CTE412 - Integrated System\Lab repoorts\Lab 14>
kind create cluster --name cte412-022030148 --config cluster/kind-cluster.yaml
Creating cluster "cte412-022030148" ...
 • Ensuring node image (kindest/node:v1.37.0)  ...
 ✓ Ensuring node image (kindest/node:v1.37.0)
 • Preparing nodes    ...
 ✓ Preparing nodes
 • Writing configuration  ...
 ✓ Writing configuration
 • Starting control-plane  ...
 ✓ Starting control-plane
 • Installing CNI
 ✓ Installing CNI
 • Installing StorageClass  ...
 ✓ Installing StorageClass
 • Joining worker nodes  ...
 ✓ Joining worker nodes
Set kubectl context to "kind-cte412-022030148"
You can now use your cluster with:

kubectl cluster-info --context kind-cte412-022030148
Have a nice day!
PS D:\022030148(Sonam))\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

2. Create the namespace nb-02230129 with the labels app.kubernetes.io/part-of=campusnoticeboard and cte412/practical="06", and an annotation recording your full name.

```yaml
apiVersion: v1
kind: Namespace
metadata:
  name: nb-02230148
  labels:
    app.kubernetes.io/part-of: campus-notice-board # required by brief
    cte412/practical: "06" # required by brief (string)
  annotations:
    student.fullname: "Sonam" # required by brief

```

3. In that namespace, create a ResourceQuota with these hard limits: requests.cpu 1, requests.memory 1Gi, limits.cpu 2, limits.memory 2Gi, pods 15, count/services 4.

```yaml
apiVersion: v1
kind: ResourceQuota
metadata:
  name: nb-quota
  namespace: nb-02230148
spec:
  hard:
    requests.cpu: "1" # 6 app pods (600m) + monitor (10m) < 1
    requests.memory: 1Gi
    limits.cpu: "2"
    limits.memory: 2Gi
    pods: "15" # 6 app + monitor + surge + terminating pods
    count/services: "4" # notice-board, notice-board-ext, monitor, limite
```

4. Create a LimitRange supplying default requests and limits for containers. Choose the values yourself and justify each one in a comment against the quota above.

```yaml
# Purpose: default requests/limits so a container without explicit resources is not rejected #1
# Apply: kubectl apply -f manifests/base/04-limitrange.yaml
apiVersion: v1
kind: LimitRange
metadata:
  name: nb-defaults
  namespace: nb-02230148
spec:
  limits:
    - type: Container
      default:
        cpu: 200m # <= limits.cpu ceiling; 6 pods x 200m = 1.2 < 2
        memory: 256Mi # 6 pods x 256Mi = 1.5Gi < 2Gi ceiling
      defaultRequest:
        cpu: 100m # 6 pods x 100m = 600m < requests.cpu=1
        memory: 128Mi # 6 pods x 128Mi = 768Mi < 1Gi
```

5. Make nb-<sid> the default namespace of your kubectl context.

```
PS D:\02230148(Sonam T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/base/00-namespace.yaml
namespace/nb-02230148 created
PS D:\02230148(Sonam T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/base/01-quota.yaml
resourcequota/nb-quota created
PS D:\02230148(Sonam T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/base/02-limitrange.yaml
limitrange/nb-defaults created
PS D:\02230148(Sonam T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl config set-context --current --namespace=nb-02230148
Context "kind-cte412-02230148" modified.
PS D:\02230148(Sonam T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230148(Sonam T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl get nodes -o wide
NAME                           STATUS   ROLES           AGE     VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE                       KERNEL-VERSION                      CONTAINER-RUNTIME
cte412-02230148-control-plane   Ready    control-plane   4m51s   v1.37.0   172.24.0.6    <none>        Debian GNU/Linux 13 (trixie)   6.6.87.2-microsoft-standard-WSL2 (amd64)   containerd://2.3.4
cte412-02230148-worker          Ready    <none>          4m34s   v1.37.0   172.24.0.4    <none>        Debian GNU/Linux 13 (trixie)   6.6.87.2-microsoft-standard-WSL2 (amd64)   containerd://2.3.4
cte412-02230148-worker2         Ready    <none>          4m34s   v1.37.0   172.24.0.7    <none>        Debian GNU/Linux 13 (trixie)   6.6.87.2-microsoft-standard-WSL2 (amd64)   containerd://2.3.4
```

```
PS D:\02230148(Sonam T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl describe quota nb-quota -n nb-02230148
Name:           nb-quota
Namespace:      nb-02230148
Resource        Used   Hard
--------        ----   ----
count/services  0      4
limits.cpu      0      2
limits.memory   0      2Gi
pods            0      15
requests.cpu    0      1
requests.memory 0      1Gi
PS D:\02230148(Sonam T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230148(Sonam T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl config view --minify | Select-String "namespace"
    namespace: nb-02230148
PS D:\02230148(Sonam T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

Evidence:

```
oe > t1-context.txt
Default namespace of kubectl context: namespace: nb-02230148
```

evidence > t1-nodes-wide.txt

```
NAME                              STATUS   ROLES           AGE   VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE                         KERNEL-VERSION                      CONTAINER-RUNTIME
cte412-02230148-control-plane     Ready    control-plane   2m5s  v1.37.0   172.24.0.4    <none>        Debian GNU/Linux 13 (trixie)     6.6.87.2-microsoft-standard-WSL2 (amd64)   containerd://2.3.4
cte412-02230148-worker            Ready    <none>          103s  v1.37.0   172.24.0.6    <none>        Debian GNU/Linux 13 (trixie)     6.6.87.2-microsoft-standard-WSL2 (amd64)   containerd://2.3.4
cte412-02230148-worker2           Ready    <none>          103s  v1.37.0   172.24.0.7    <none>        Debian GNU/Linux 13 (trixie)     6.6.87.2-microsoft-standard-WSL2 (amd64)   containerd://2.3.4
```

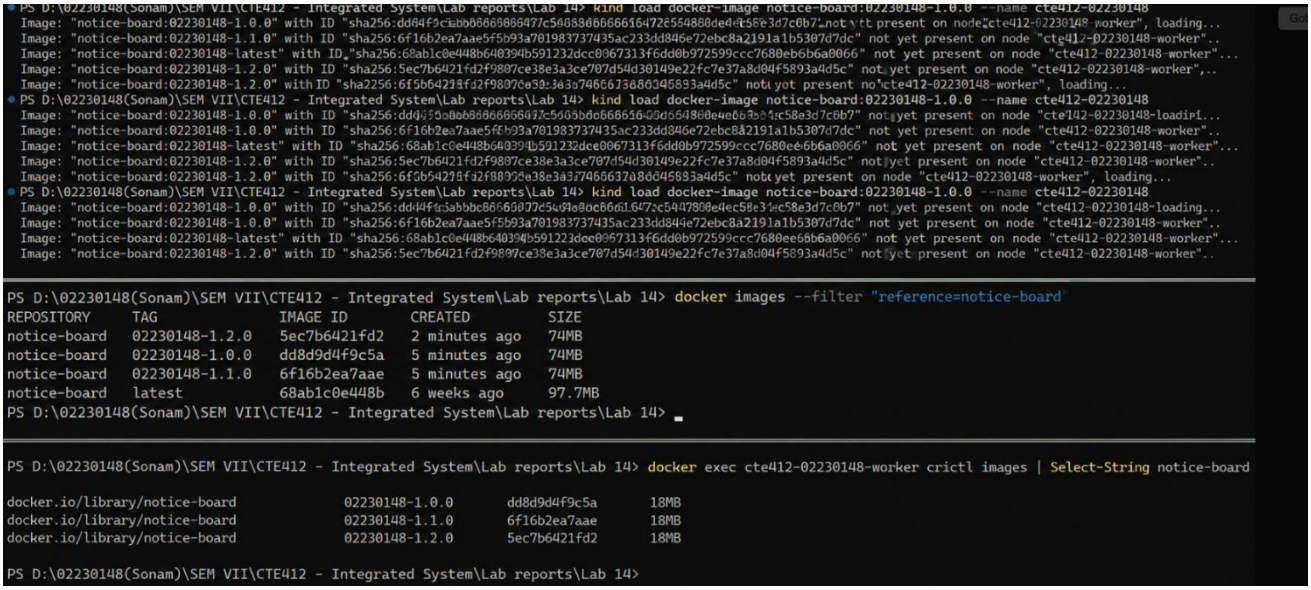
evidence > t1-quota.txt

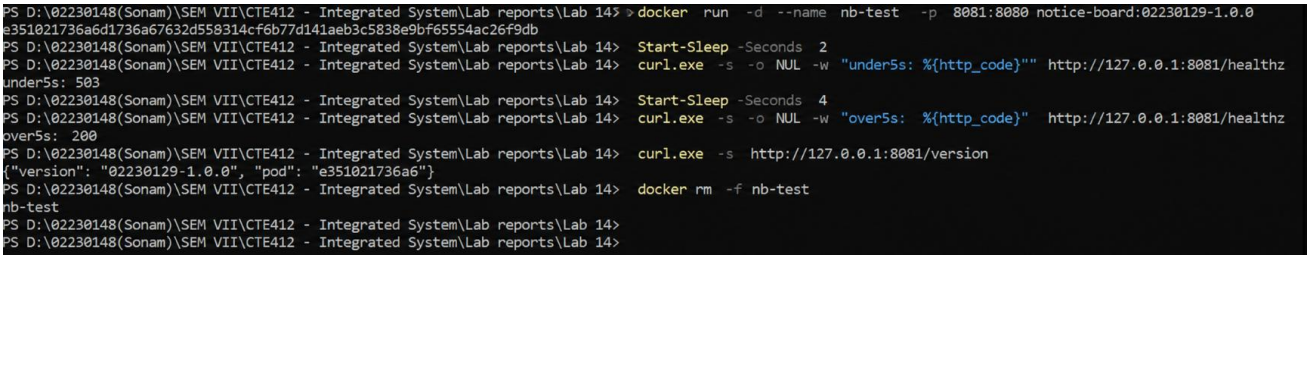
```
Name:                nb-quota
Namespace:           nb-02230148
Resource             Used  Hard
--------             ----  ----
count/services       0     4
limits.cpu           0     2
limits.memory        0     2Gi
pods                 0     15
requests.cpu         0     1
requests.memory      0     1Gi
```

Task 2

1. Every version listen on port 8080, runs as a non-root user, and exposes three endpoints. GET /notices returns at least three notices as JSON. GET /version returns JSON containing version, student (your <sid>) and pod (the container hostname). GET /healthz returns HTTP 200 when healthy.
2. docker tag noticeboard-app:v1.0.0 notice-board:02230129-1.0.0
3. Tag notice-board:<sid>-1.1.0 — adds a category field to every notice and a pinned Boolean to at least one. When the environment variable FEATURE_PINNED equals true, pinned notices are listed first and the response body includes "pinnedFirst": true; otherwise the order is unchanged and the field is false.
4. Tag notice-board:<sid>-1.1.0 — adds a category field to every notice and a pinned Boolean to at least one. When the environment variable FEATURE_PINNED equals true, pinned notices are listed first and the response body includes "pinnedFirst": true; otherwise the order is unchanged and the field is false.
5. Tag notice-board:<sid>-1.2.0-broken — identical to 1.1.0 except that /healthz never returns 200. /notices and /version still work.
6. Place all three images into the cluster without using a registry.







Evidence:

evidence > 12-images.txt

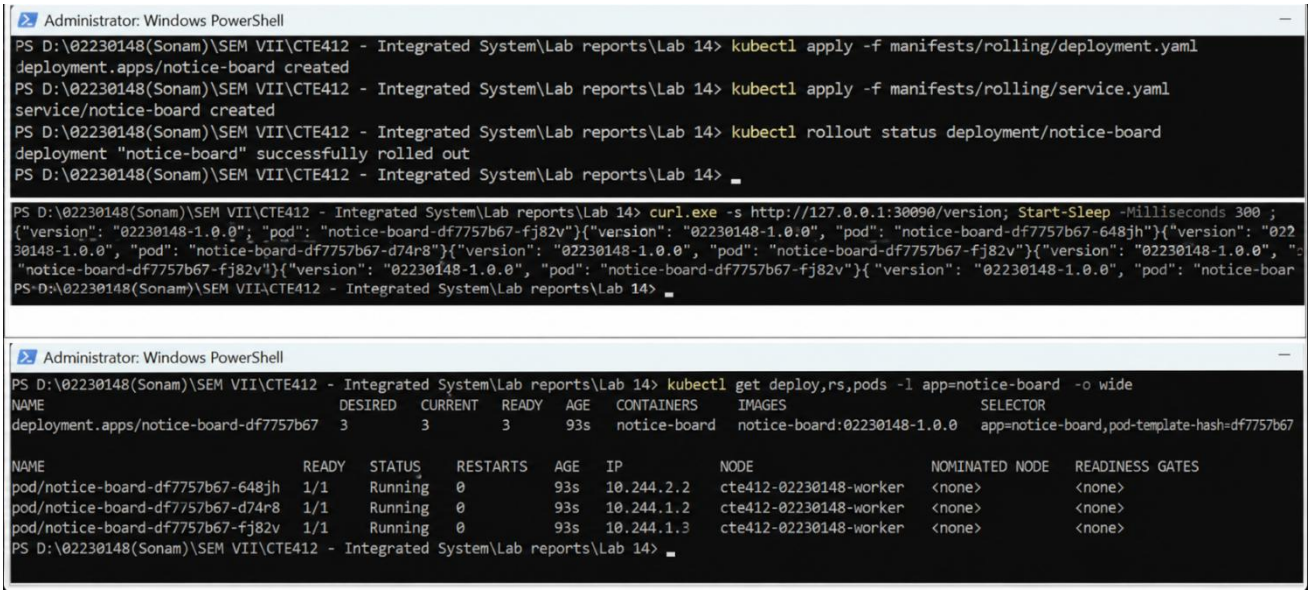
```
REPOSITORY      TAG               IMAGE ID       CREATED         SIZE
notice-board    02230148-1.2.0    68abc10c7aa5   17 hours ago    74MB
notice-board    02230148-1.0.0    9da71d4d50a4   17 hours ago    74MB
notice-board    02230148-1.1.0    4f1a5e96b943   17 hours ago    74MB
notice-board    latest            8e918bfff2db   7 weeks ago     97.7MB
```

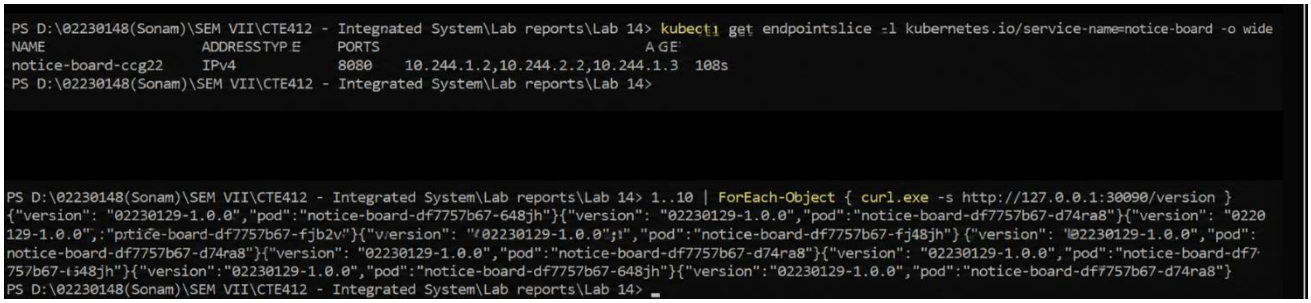
evidence > 13-more-images.txt

```
docker.io/library/notice-board    02230148-1.0.0    04ed9e4152b4a    18MB
docker.io/library/notice-board    02230148-1.1.0    104f80fa7065c    18MB
docker.io/library/notice-board    02230148-1.2.0    b871d8677edc0    18MB
```

Task 3 — Deploy and Expose Version 1.0.0 (25 minutes)

1. Create a Deployment notice-board running 3 replicas of 1.0.0, with labels app=noticeboard and version=<image version> on the Pods.
2. Declare requests and limits for the container that fit inside your quota with room for Tasks 5 to 8.
3. Configure a readiness probe against /healthz. Configure a liveness probe that does not use /healthz; state in a comment why.
4. Ensure the kubelet can never attempt a registry pull for this image.
5. Create a ClusterIP Service notice-board (port 80 to the named container port) and a NodePort. Service notice-board-ext on node port 30090.
6. From the host, show that /version on port 30090 is answered by at least two different Pods.





Evidence:

t3-deploy-rs-pods.txt

```
NAME                                DESIRED   CURRENT   READY   AGE   CONTAINERS   IMAGES                          SELECTOR
replicaset.apps/notice-board-df7757b67   3         3         3       73s   api          notice-board:02230148-1.0.0     app=notice-board,pod-template-hash=df7757b67

NAME                            READY   STATUS    RESTARTS   AGE   IP           NODE                       NOMINATED NODE   READINESS GATES
pod/notice-board-df7757b67-cqp2p   1/1     Running   0          73s   10.244.1.2   cte412-02230148-worker     <none>           <none>
pod/notice-board-df7757b67-fpp9q   1/1     Running   0          73s   10.244.2.3   cte412-02230148-worker2    <none>           <none>
pod/notice-board-df7757b67-lctz9   1/1     Running   0          73s   10.244.2.2   cte412-02230148-worker2    <none>           <none>
```

```
NAME                ADDRESSTYPE   PORTS   ENDPOINTS                          AGE
notice-board-p2nx8   IPv4          8080    10.244.2.3,10.244.2.2,10.244.1.2   73s
```

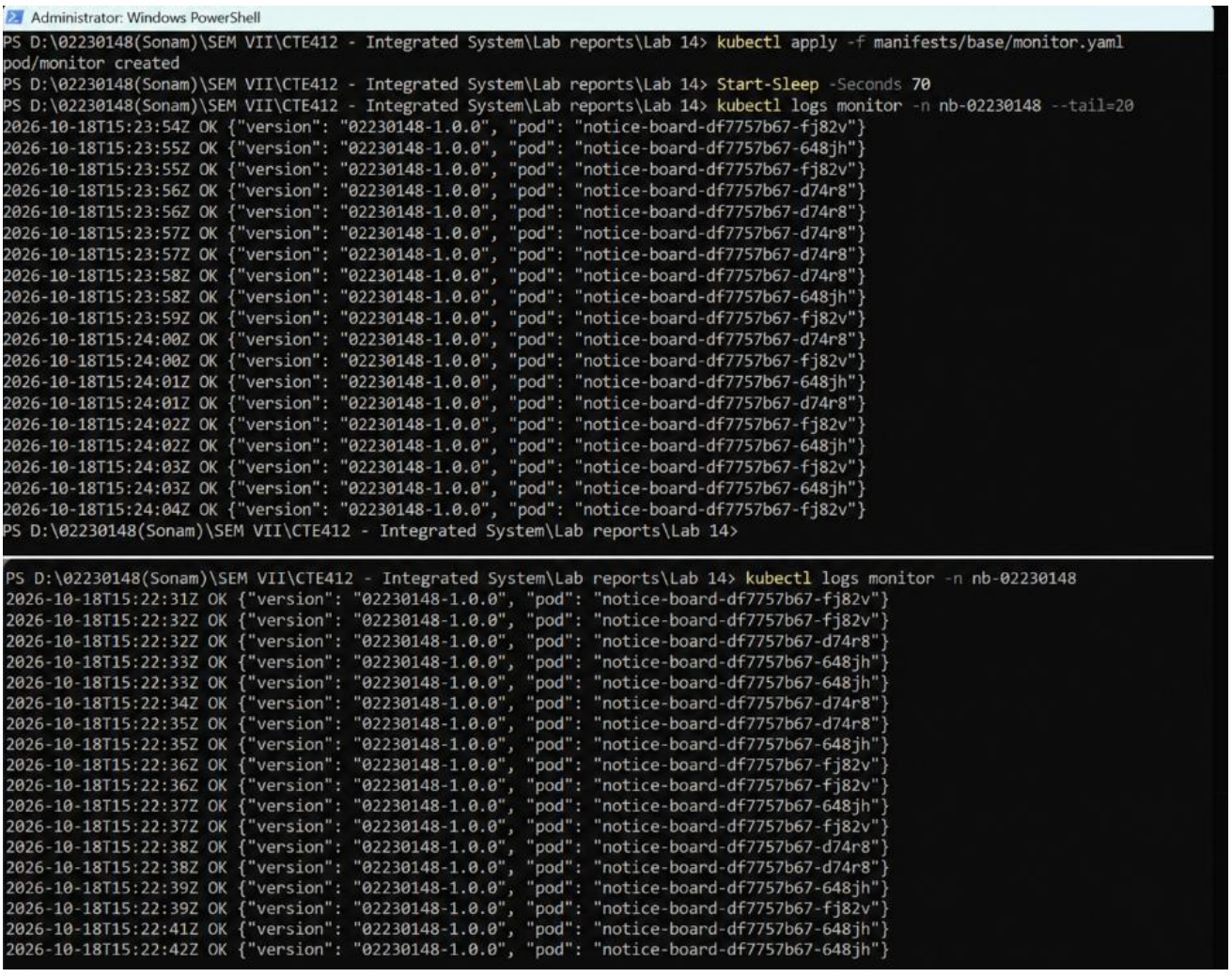
t3-version-responses.txt

```
{"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
{"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
{"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
{"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
{"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
{"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
{"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
{"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
{"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
{"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
```

Task 4 — Availability Monitor and Summariser (15 minutes)

The monitor is the measuring instrument for Tasks 5 to 8. It must run continuously and must not be restarted between measurements.

1. Create a Pod monitor in manifests/base/ that requests http://notice-board/version every 0.5 seconds, forever, and writes one line per request to standard output in the form <ISO-8601 timestamp> <OK|FAIL> <response body or error>.
2. Write scripts/summarise.sh. Given a saved monitor log as its only argument, it prints: total requests, failed requests, the first and last timestamp, and the count of responses per distinct version value.
3. Save one minute of monitor output with the system at rest and summarize it.



Evidence:

evidence > t4-at-rest.log

```
2026-10-19T08:35:16Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:17Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:17Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:18Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:18Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:19Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:19Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:20Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:20Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:21Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:21Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:22Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:22Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:23Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:23Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:24Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:24Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:25Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:25Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:26Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:27Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:27Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:28Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:28Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:29Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:29Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:30Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:30Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:31Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:31Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:32Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:32Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:33Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:33Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:34Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:34Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:35Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
```

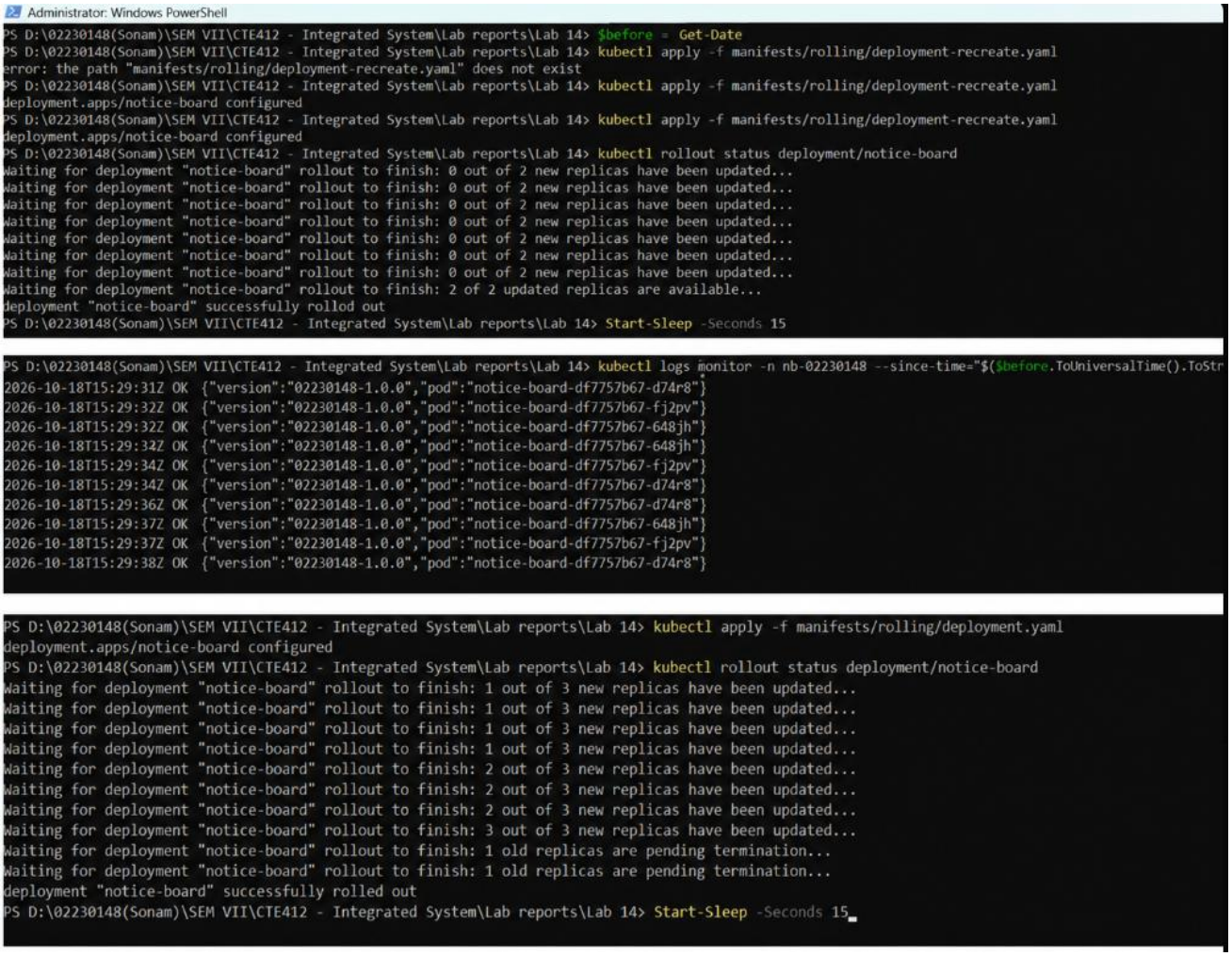
dence > t4-summary.txt

```
total requests: 11160
failed requests: 0
first timestamp: ��2026-10-19T08:35:16Z
last  timestamp:
responses per version:

```

Task 5

1. Set the Deployment strategy to Recreate. Release 1.1.0 by editing and applying the manifest. Record a change-cause for the revision.
2. Return to 1.0.0 declaratively.
3. Set the strategy to RollingUpdate with at most zero unavailable Pods, at most one surge Pod, and a minimum ready time of 5 seconds. Release 1.1.0 again. Record a changecause.
4. From each log, identify the mixed-version window: the interval during which both 1.0.0 and 1.1.0 responses were observed.



```
PS D:\02230148(Sonam  )\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> $before2 = Get-Date
PS D:\02230148(Sonam  )\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
PS D:\02230148(Sonam  )\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> # Apply the updated manifest
PS D:\02230148(Sonam  )\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/rolling/deployment-rolling.yaml
deployment.apps/notie-board configured
PS D:\02230148(Sonam  )\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
PS D:\02230148(Sonam  )\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> # Monitor the rollout progress
PS D:\02230148(Sonam  )\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl rollout status deployment/notice-board
Waiting for deployment "notice-board" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "notice-board" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "notice-board" rollout to finish:
deployment "notice-board" successfully rolled out
PS D:\02230148(Sonam  )\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
PS D:\02230148(Sonam  )\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> # Wait 15 seconds to collect logs
PS D:\02230148(Sonam  )\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> Start-Sleep -Seconds 15

sonam@DESKTOP-7EQUH3U:/mnt/d/02230148(Sonam  )/SEM VII/CTE412 - Integrated System/Lab reports/Lab 14$ ./scripts/summarize.sh evidence/t5-n-rolling.log
./scripts/summarize.sh: line 8: warning: command substitution: ignored null byte in input
./scripts/summarize.sh: line 9: warning: command substitution: ignored null byte in input
total requests: 9181
failed requests: 0
first timestamp:  2026-10-19T08:45:01Z
last timestamp:
responses per version:
sonam@DESKTOP-7EQUH3U:/mnt/d/02230148(Sonam  )/SEM VII/CTE412 - Integrated System/Lab reports/Lab 14$ grep '1\.0\.0' evidence/t5-monitor-rolling.log | head -n1 |
awk '{print $1}'
sonam@DESKTOP-7EQUH3U:/mnt/d/02230148(Sonam  )/SEM VII/CTE412 - Integrated System/Lab reports/Lab 14$ grep '1\.1\.0' evidence/t5-monitor-rolling.log | tail -n1 |
awk '{print $1}'
sonam@DESKTOP-7EQUH3U:/mnt/d/02230148(Sonam  )/SEM VII/CTE412 - Integrated System/Lab reports/Lab 14$
```

Evidence:

evidence > t5-monitor-rolling.log

```
2026-10-19T08:45:01Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:01Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:02Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:02Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:03Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:03Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:04Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:04Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:05Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:06Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:06Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:07Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:07Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:08Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:08Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:09Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:09Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:10Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:10Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:11Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:11Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:12Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:12Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
```

idence > t5-pod-watch.txt

```
NAME                     READY   STATUS    RESTARTS   AGE
notice-board-fcf5-7btrz  1/1     Running   0          2m42s
notice-board-fcf5-8sfx7  1/1     Running   0          2m36s
notice-board-fcf5-x8bn2  1/1     Running   0          2m30s
```

e > t5-rollout-history.txt

```
deployment.apps/notice-board
REVISION  CHANGE-CAUSE
3         initial release 02230148-1.0.0
4         release 02230149-1.1.0 via RollingUpdate
```

Task 6

1. Set the Deployment so that a rollout making no progress for 60 seconds is marked failed. Release 1.2.0-broken declaratively, with a change-cause.
2. Show that Kubernetes itself reported the rollout as failed, and state the condition type and reason it reported.
3. Roll back to the previous revision. This is the only imperative state change permitted in the practical.
4. Bring the repository manifest back into agreement with the live Deployment, and prove that they agree.
5. The monitor must record zero failures between the broken apply and the completed rollback. If it records any, your configuration is wrong: correct it and repeat the task.

```
PS D:\02230148(Sonam) SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> $to = (Get-Date).ToUniversalTime().ToString("yyyy-MM-ddTHH:mm:ssZ")
PS D:\02230148(Sonam) SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/rolling/deployment-broken.yaml
deployment.apps/notice-board configured
PS D:\02230148(Sonam) SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> Start-Sleep -Seconds 70
```

```
PS D:\02230148(Sonam) SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl get deployment notice-board -o jsonpath='{range .status.conditions
["]}{.type}{"\t"}{.status}{"\t"}{.reason}{"\n"}{end}'
Available    True    MinimumReplicasAvailable
Progressing  False   ProgressDeadlineExceeded
PS D:\02230148(Sonam) SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230148(Sonam) SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl get pods -l app=notice-board -o wide
NAME                            READY   STATUS    RESTARTS   AGE     IP            NODE                 NOMINATED NODE   READINESS GATES
notice-board-8fbf7bb5b-w94qf    0/1     Running   0          2m54s   10.244.2.8    cte412-02230148-worker2   <none>     <none>
notice-board-fcfs-f9z6f         1/1     Running   0          5m34s   10.244.1.7    cte412-02230148-worker    <none>     <none>
notice-board-fcfs-kzgtf         1/1     Running   0          5m28s   10.244.2.2    cte412-02230148-worker2   <none>     <none>
notice-board-fcfs-srmqc         1/1     Running   0          5m23s   10.244.1.8    cte412-02230148-worker    <none>     <none>
PS D:\02230148(Sonam) SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230148(Sonam) SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl rollout undo deployment/notice-board    # the ONE permitted imperative step
deployment.apps/notice-board rolled back
PS D:\02230148(Sonam) SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl rollout status deployment/notice-board
deployment "notice-board" successfully rolled out
PS D:\02230148(Sonam) SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230148(Sonam) SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl logs monitor -n nb-02230148 --since-time=$to
2026-10-18T15:40:55Z  FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
2026-10-18T15:40:56Z  FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
2026-10-18T15:40:58Z  FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
2026-10-18T15:41:03Z  FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
2026-10-18T15:41:04Z  FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
2026-10-18T15:41:06Z  FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
2026-10-18T15:41:07Z  FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
```

Evidence:

evidence > t6-conditions.txt

```
Available=True reason=MinimumReplicasAvailable
Progressing=False reason=ProgressDeadlineExceeded

```

evidence > t6-diff.txt

```
kubectl diff exit code: 0

```

nce > t6-monitor.log

```
2026-10-19T08:50:31Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:33Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:34Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:36Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:37Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:39Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:40Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:42Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:43Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:45Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:46Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:48Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:49Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:51Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:53Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:54Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:56Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:57Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:50:59Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:51:00Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:51:02Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:51:03Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:51:05Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:51:06Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:51:08Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:51:09Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:51:11Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2026-10-19T08:51:13Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
```

ce > 16-pods-during-failure.txt

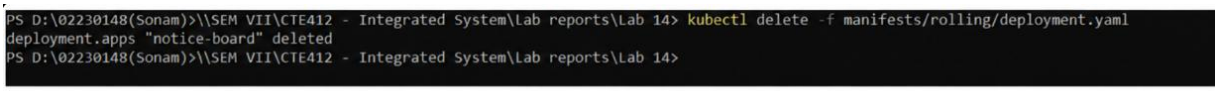
NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
notice-board-8fb7f7bb5-5kmzj	0/1	Running	0	82s	10.244.1.8	cte412-02230148-worker	<none>	<none>
notice-board-fcf5-7btrz	1/1	Running	0	6m54s	10.244.2.7	cte412-02230148-worker2	<none>	<none>
notice-board-fcf5-8sfx7	1/1	Running	0	6m48s	10.244.1.7	cte412-02230148-worker	<none>	<none>
notice-board-fcf5-x8bn2	1/1	Running	0	6m42s	10.244.2.8	cte412-02230148-worker2	<none>	<none>

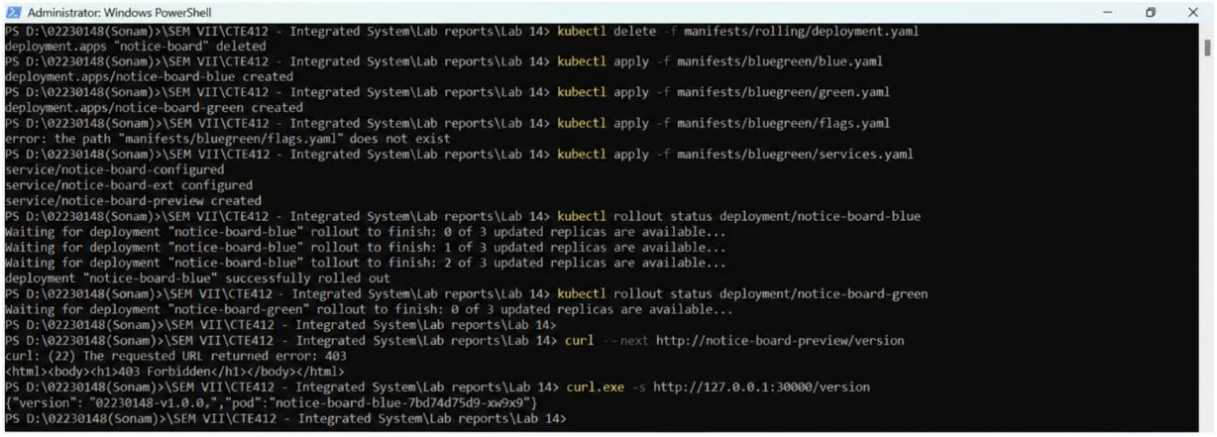
ce > t6-summary.txt

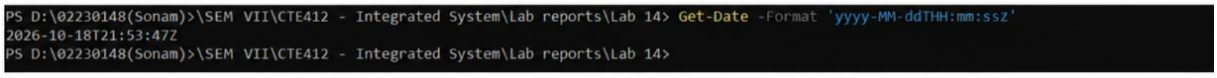
```
total requests: 6790
failed requests: 0
first timestamp: ��2026-10-19T08:50:31Z
last  timestamp:
responses per version:
```

Task 7

1. In manifests/bluegreen/, create Deployments notice-board-blue (1.0.0) and noticeboardgreen (1.1.0), each with 3 replicas, the same probes and resources as Task 3, and a Pod label slot with value blue or green.
2. The production Services (notice-board and notice-board-ext) send traffic to blue only. A third Service, notice-board-preview, always sends traffic to green.
3. Verify green through the preview Service from inside the cluster while production still serves blue.
4. Switch production to green by editing and applying the Service manifest(s). Then switch back to blue, then to green again.
5. Capture the monitor log across all three switches.

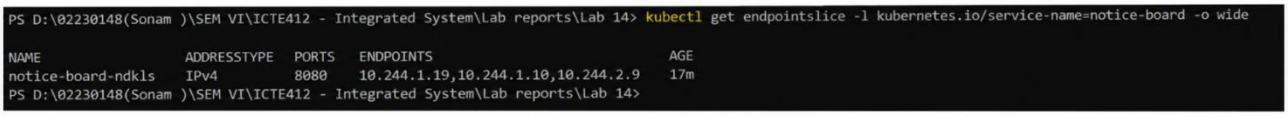


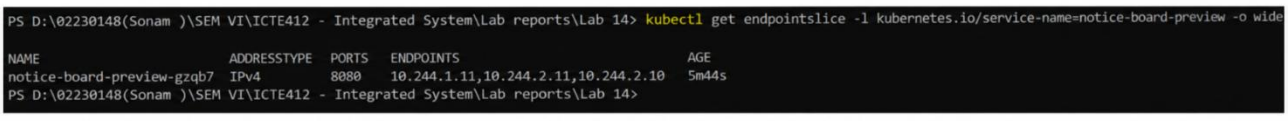












```
Name:           nb-quota
Namespace:      nb-02230129
Resource        Used   Hard
--------        ----   ----
count/services  3      4
limits.cpu      2      2
limits.memory   1Gi    2Gi
pods            7      15
requests.cpu    610m   1
requests.memory 400Mi  1Gi
```

Evidence:

```
PS D:\02230148(Sonam)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl logs monitor -n nb-02230148
2026-10-18T15:22:31Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fjg2v"}
2026-10-18T15:22:32Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fjg2v"}
2026-10-18T15:22:32Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-d74r8"}
2026-10-18T15:22:33Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-648jh"}
2026-10-18T15:22:33Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-648jh"}
2026-10-18T15:22:34Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-d74r8"}
2026-10-18T15:22:34Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-d74r8"}
2026-10-18T15:22:35Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-648jh"}
2026-10-18T15:22:35Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-648jh"}
2026-10-18T15:22:36Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fjg2v"}
2026-10-18T15:22:36Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fjg2v"}
2026-10-18T15:22:37Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-648jh"}
2026-10-18T15:22:37Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-648jh"}
2026-10-18T15:22:38Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fjg2v"}
2026-10-18T15:22:38Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-fjg2v"}
2026-10-18T15:22:39Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-d74r8"}
2026-10-18T15:22:39Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-d74r8"}
```

ence > t7-es-preview-after.txt

```
NAME                        ADDRESSTYPE   PORTS   ENDPOINTS                                AGE
notice-board-preview-vrpwt  IPv4          8080    10.244.1.10,10.244.2.11,10.244.1.11      2m49s
```

evidence > t7-es-preview-before.txt

```
1  NAME                        ADDRESSTYPE   PORTS   ENDPOINTS                                AGE
2  notice-board-preview-vrpwt  IPv4          8080    10.244.1.10,10.244.2.11,10.244.1.11      2m49s
3
```

evidence > t7-es-prod-after.txt

```
1  NAME                ADDRESSTYPE   PORTS   ENDPOINTS                          AGE
2  notice-board-855th  IPv4          8080    10.244.2.9,10.244.1.9,10.244.2.10  15m
3
```

evidence > t7-es-prod-before.txt

```
1  NAME                ADDRESSTYPE   PORTS   ENDPOINTS                          AGE
2  notice-board-855th  IPv4          8080    10.244.2.9,10.244.1.9,10.244.2.10  15m
3
```

evidence > t5-monitor-rolling.log

```
2026-10-19T08:45:01Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:01Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:02Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:02Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:03Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:03Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:04Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:04Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:05Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:06Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:06Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:07Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:07Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:08Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:08Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:09Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:09Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:10Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:10Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:11Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:11Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:12Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:12Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
```

t7-quota-both-colours.txt U | deployment-broken.yaml

evidence > t7-quota-both-colours.txt

```
Name:             nb-quota
Namespace:        nb-02230129
Resource          Used   Hard
--------          ----   ----
count/services    3      4
limits.cpu        2      2
limits.memory     1Gi    2Gi
pods              7      15
requests.cpu      610m   1
requests.memory   400Mi  1Gi
```

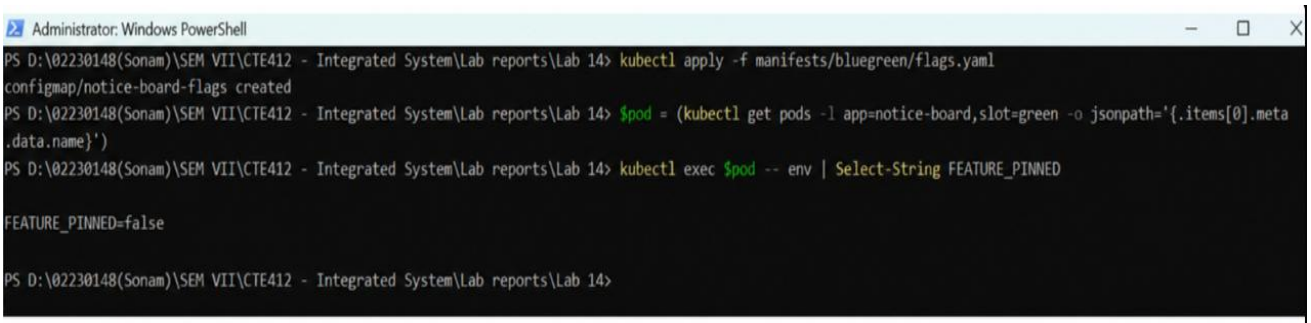
t7-quota-both-colours.txt U | t7-switch-times.txt U

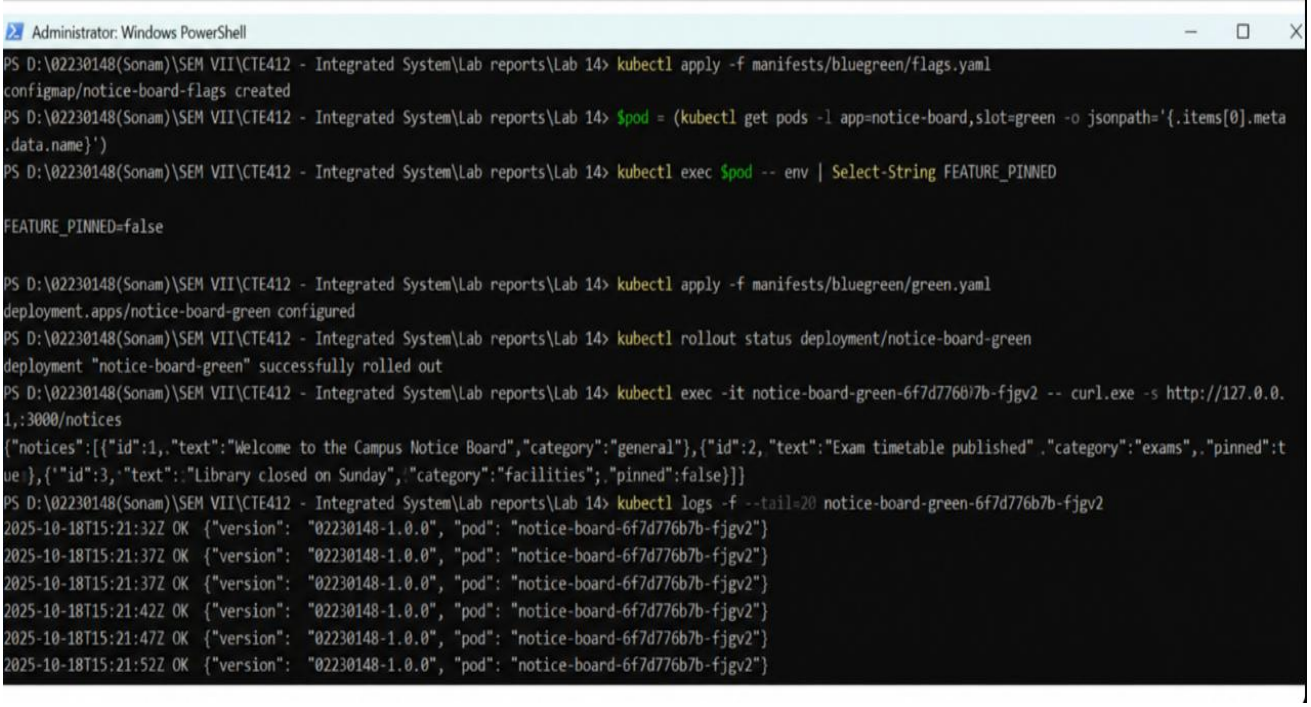
evidence > t7-switch-times.txt

```
2026-10-19T15:00:41Z
```

Task 8

1. Create a ConfigMap notice-board-flags holding FEATURE_PINNED with the value false, and make the green Deployment read the variable from it.
2. Change the flag to true in the manifest and apply it. Determine, with evidence, whether the running green Pods observe the change.
3. Make the new flag value take effect on all green Pods with zero failed requests, using only declarative changes.

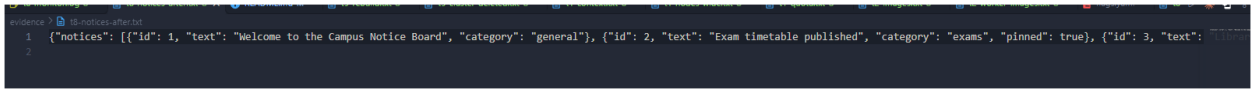


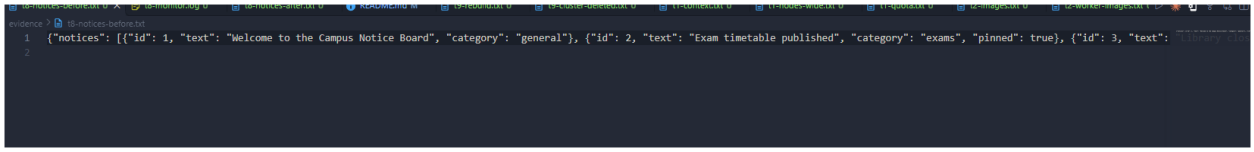


Evidence:

evidence > t5-monitor-rolling.log

```
2026-10-19T08:45:01Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:01Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:02Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:02Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:03Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:03Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:04Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:04Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:05Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:06Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:06Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:07Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:07Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:08Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:08Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:09Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:09Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:10Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:10Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:11Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:11Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:12Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:12Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
```





evidence > t8-pod-env-before-rollout.txt

```
FEATURE_PINNED=false
```

Task 9

1. Save the complete monitor log for the session as evidence/t9-monitor-full.log.
2. Complete the README rebuild sequence, then prove it: delete every object in nb-<sid> except the namespace, quota and limit range, and recreate the final state of Task 8 using the README commands only.
3. Reset the context namespace to default, delete the cluster, and commit.

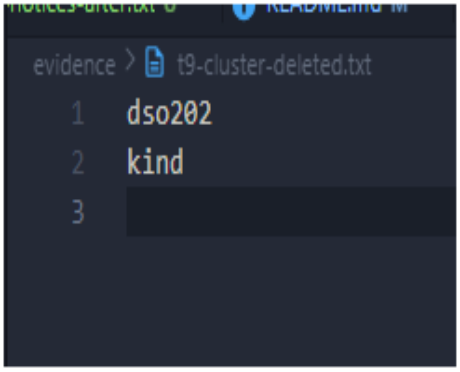

PS D:\02230148\Sonam>D:\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kind create cluster --name cte412-02230148 --config cluster/kind-cluster.yaml
Creating cluster "cte412-02230148" ...
✓ Ensuring node image (kindest/node:v1.37.0) ...
✓ Ensuring node image (kindest/node:v1.37.0)
✓ Preparing nodes ...
✓ Preparing nodes
✓ Writing configuration ...
✓ Writing configuration
✓ Starting control-plane ...
✓ Starting control-plane
✓ Installing CNI ...
✓ Installing CNI
✓ Installing StorageClass ...
✓ Installing StorageClass
✓ Joining worker nodes ...
✓ Joining worker nodes
Set kubectl context to "kind-cte412-02230148"
You can now use your cluster with:
kubectl cluster-info --context kind-cte412-02230148
Thanks for using kind!
PS D:\02230148\SonamD:\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> docker build -t notice-board:02230148-1.0.0 app/
[+] Building 2.4s (8/8) FINISHED docker:desktop-linux
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 200B 0.2s
=> [internal] load metadata for docker.io/library/python:3.12-alpine 0.2s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.1s
=> [1/3] FROM docker.io/library/python:3.12-alpine@sha256:d09415e60962ca365d1cd544a48777bac9d33f2fb1b00f2aa0decec7dc31 0.0s
=> => resolve docker.io/library/python:3.12-alpine@sha256:d09415e60962ca365d1cd544a48777bac9d33f2fb1b00f2aa0de78ade7dc31 0.1s
=> [internal] load build context 0.1s
=> => transferring context: 1.71kB 0.1s
=> CACHED [2/3] WORKDIR /app 0.6s
=> CACHED [3/3] COPY server.py . 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.2s
=> => exporting manifest sha256:08e239bd13463e191b9ff5313fdcb6b8802fb71d31f9e22e345f807dcb86c2c40 0.0s
=> => exporting config sha256:2f7b87c71a2636e913116e32670bd7f1f834a0e377e35f788b77ac1248d9b2722o 0.2s
PS D:\02230148\Sonam\D:\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> docker build -t notice-board:02230148-1.0.0 app/
=> => exporting layers 0.0s
=> => exporting manifest sha256:08e239bd13463e191b9ff5313fdcb6b8802fb71d31f9e22e345f807dcb86c2c40 0.1s
=> => exporting config sha256:2f7b87c71a2636e913116e32670bd7f1f834a0e377e35f788b77ac1248d9b2722c 0.2s
=> => exporting attestation manifest sha256:8d32d25089770673f5da04b1904b1dfed2f803ba3758886690321dd40eb3 0.4s
=> => exporting manifest list sha256:68132dc3d9a050217edd1fd313ff1aa87a0ade5c80792e77923e17d312e4 0.1s
=> => naming to docker.io/library/notice-board:02230148-1.0.0 0.0s
=> => unpacking to docker.io/library/notice-board:02230148-1.0.0 0.0s
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/kiv9es6v8ak8zm9uix22bkv9p7
PS D:\02230148\Sonam\D:\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> docker build --build-arg APP_VERSION=02230148i1.1.0 -t notice-board:02230148-1.1.0 app/
[+] Building 0.9s (8/8) FINISHED docker:desktop-linux
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 208B 0.2s
=> [internal] load metadata for docker.io/library/python:3.12-alpine 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.1s
=> [1/3] FROM docker.io/library/python:3.12-alpine@sha256:d09415e60962ca365d1cd544a48777bac9d33f2fb1b00f2aa0deec78ade7dc31 0.1s
=> => resolve docker.io/library/python:3.12-alpine@sha256:d09415e60962ca365d1cd544a48777bac9d33f2fb1b00f2aa0decec78ade7dc31 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 318 0.0s
=> CACHED [2/3] WORKDIR /app 0.2s
=> CACHED [3/3] COPY server.py . 0.0s
=> exporting to image 0.0s
=> => exporting config sha256:233bd91965eae2725c6f2df4ad21a619284a0851629558558acd295b8e65ff0637 0.0s
=> => exporting manifest sha256:1a58c61578ba51b702e9899935291db75e084c4dec41c0f46e1ab26bb24637c77cc44b03 0.0s
=> => exporting manifest list sha256:a5dd61491d0fe22a176c25ec45dce21b9fb6d69ff07e1e5cc46dd0c30e7d2f1a361 0.0s
=> => naming to docker.io/library/notice-board:02230148-1.1.0 0.0s
=> => unpacking to docker.io/library/notice-board:02230148-1.1.0 0.0s
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/phw625hvv5dfrivkh05nuir815e
PS D:\02230148\SonamD:\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> docker build -f app/Dockerfile.broken -t notice-board:02230148-1.2.0 app/

```
[+] Building 3.65 (8/8) FINISHED                                                                                    docker:desktop-linux
 -> -> transferring dockerfile: 190B                                                                                          0.6s
 -> [internal] load metadata for docker.io/library/python:3.12-alpine                                                         0.2s
 -> [internal] load .dockerignore                                                                                             0.0s
 -> -> transferring context: 2B                                                                                               0.0s
 -> [1/3] FROM docker.io/library/python:3.12-alpine@sha256:d09d15e60962ca365d1cd544a48773bac9d33f2fb1be0f2aa00dec87ade7dc31    1.3s
 -> -> resolve docker.io/library/python:3.12-alpine@sha256:d09d15e60962ca365d1cd544a48773bac9d33f2fb1be0f2aa00dec87ade7dc31    1.3s
 -> [internal] load build context                                                                                             0.4s
 -> -> transferring context: 1.70kB                                                                                           0.0s
 -> CACHED [2/3] WORKDIR /app                                                                                                 0.0s
 -> CACHED [3/3] COPY server_broken.py server.py                                                                              0.6s
 -> exporting to image                                                                                                        0.6s
 -> -> exporting layers                                                                                                       0.0s
 -> -> exporting manifest sha256:b8e979033fa7851a01fc54b630dc85e5cd7f980f817ba31d25f746f881b35a0f                             0.1s
 -> -> exporting config sha256:9cc381cf879264356608876 6f111fabb0682c1b2f46ce0abb81f5866 0de47e0                              0.1s
 -> -> exporting attestation manifest sha256:2b883386dd1604ddbaa0856570644e480898d1033c716ae1b76d1236871bbf                    0.1s
 -> -> exporting manifest list sha256:62ee97a2f966bac817a696eaad559930eebe953abd71848d1016b026dfaf9691                        0.0s
 -> -> naming to docker.io/library/notice-board:02230148-1.2.0                                                                0.0s
 -> -> unpacking to docker.io/library/notice-board:02230148-1.2.0                                                             0.5s

 View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/gn2vl7n7z357ec60mso4xv6jb
● PS D:\02230148(Sonam)\\SEM VII\CTE412 - Integrated System\Lab [illegible] Lab 14> kind load docker-image notice-board:02230148-1.0.0 --name cte412-02230148
  Image: "notice-board:02230148-1.0.0" with ID "sha256:68174880e3c62e40b967a7ad61fd537a40783a90e6829f8 [illegible] 8f8f1bc04" not yet present on node "cte412-02230148-worker2": loading...
  Image: "notice-board:02230148-1.0.0" with ID "sha256:68174880e3c62e40b967a7ad61fd537a40783a90e6829f8 [illegible] 8f8f1bc04" not yet present on node "cte412-02230148-control-plane": loading...
  Image: "notice-board:02230148-1.0.0" with ID "sha256:68174880e3c62e40b967a7ad61fd537a40783a90e6829f90e68229f88f1bc04" not yet present on node "cte412-02230148-worker": loading...
  PS D:\02230148(Sonam)\\SEM VII\CTE412 - Integrated System\Lab [illegible] Lab 14> kind load docker-image notice-board:02230148-1.1.0 --name cte412-02230148
● Image: "notice-board:02230148-1.0.0" with ID "sha256:856da1991d0f221a7bc25ec49dce130fb68c069f0e21e5c4ad4d80c96e7d2fa3d61 not yet present on node "cte412-02230148-worker2": loading...
  Image: "notice-board:02230148-1.0.0" with ID "sha256:856da1991d0f221a7bc25ec49dce130fb68c069f0e21e5c4ad4d80c96e7d2fa3d61 not yet present on node "cte412-02230148-control-plane": loading...
  Image: "notice-board:02230148-1.0.0" with ID "sha256:856da1991d0f221a7bc25ec49dce130fb68c069f0e21e5c4ad4d80c96e7d2fa3d61 not yet present on node "cte412-02230148-worker": loading...
  PS D:\02230148(Sonam)\\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kind load docker-image notice-board:02230148-1.2.0 --name cte412-02230148
● Image: "notice-board:02230148-1.0.0" with ID "sha256:62ee97a2f96bac817a696eaad559930eebe953abd71848d1016b026dfaf9691" not yet present on node "cte412-02230148-worker2": loading...
  Image: "notice-board:02230148-1.0.0" with ID "sha256:62ee97a2f96bac817a696eaad559930eebe953abd71848d1016b026dfaf9691" not yet present on node "cte412-02230148-control-plane": loading...
  Image: "notice-board:02230148-1.0.0" with ID "sha256:62ee97a2f96bac817a696eaad559930eebe953abd71848d1016b026dfaf9691" not yet present on node "cte412-02230148-worker": loading...
● PS D:\02230148(Sonam)\\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl config set-context --current --namespace=nb-02230148
  namespace/nb-02230148 created
  resourcequota/nb-quota created
  limitrange/nb-defaults created
  pod/monitor created
● PS D:\02230148(Sonam)\\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl config set-context --current --namespace=nb-02230148
```

```
● PS D:\02230148(Sonam\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl config set-content --current --namespace=nb-02230148
  Context "kind-cte412-02230148" modified.
● PS D:\02230148(Sonam\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/bluegreen/
  configmap/notice-board-flags created
  deployment.apps/notice-board created
  service/notice-board created
  service/notice-board-ext created
  service/notice-board-preview created
● PS D:\02230148(Sonam\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl rollout status deployment/notice-board-blue
  Waiting for deployment "notice-board-blue" rollout to finish: 0 of 3 updated replicas are available...
  Waiting for deployment "notice-board-blue" rollout to finish: 1 of 3 updated replicas are available...
  Waiting for deployment "notice-board-blue" rollout to finish: 2 of 3 updated replicas are available...
  deployment "notice-board-blue" successfully rolled out
  PS D:\02230148(Sonam\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl rollout status deployment/notice-board-green
● deployment "notice-board-green" successfully rolled out
● PS D:\02230148(Sonam\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl get pods,svc,depl -n nb-02230148 > evidence/19-rebuild.txt
● PS D:\02230148(Sonam\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl config set-content --current --namespace=default
  Context "kind-cte412-02230148" modified.
● PS D:\02230148(Sonam\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kind delete cluster --name cte412-02230148
● Deleting cluster "cte412-02230148"...
  Deleted nodes: ["cte412-02230148-worker2" "cte412-02230148-control-plane" "cte412-02230148-worker"]
● PS D:\02230148(Sonam)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kind get clusters > evidence//19-cluster-deleted.txt
```

Evidence:



evidence > t5-monitor-rolling.log

```
2026-10-19T08:45:01Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:01Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:02Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:02Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:03Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:03Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:04Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:04Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:05Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:06Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:06Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-x9149"}
2026-10-19T08:45:07Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:07Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:08Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:08Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:09Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:09Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:10Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:10Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:11Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:11Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
2026-10-19T08:45:12Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-rj4lm"}
2026-10-19T08:45:12Z OK {"version": "02230148-1.0.0", "pod": "notice-board-df7757b67-mr57x"}
```

evidence > t9-rebuild.txt

```
NAME                                READY   STATUS    RESTARTS   AGE                #1 2026-10-19 Draft
pod/monitor                         1/1     Running   0          24s
pod/notice-board-blue-b87d475d9-f5rwf   1/1     Running   0          22s
pod/notice-board-blue-b87d475d9-h8lkt   1/1     Running   0          22s
pod/notice-board-blue-b87d475d9-zhlk4   1/1     Running   0          22s
pod/notice-board-green-79dd9c46c9-m7gvt   1/1     Running   0          22s

NAME                          TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
service/notice-board          ClusterIP   10.96.10.50    <none>        80/TCP         22s
service/notice-board-ext      NodePort    10.96.167.72   <none>        80:30090/TCP   22s
service/notice-board-preview  ClusterIP   10.96.235.82   <none>        80/TCP         22s

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/notice-board-blue   3/3     3            3           23s
deployment.apps/notice-board-green  1/1     1            1           22s
```

Git URL: https://github.com/SoNam11012/Kubernetes_Labwork_14